Title of paper

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Abstract

Example abstract for the High Energy Astrophysics Journal. Here you provide a brief summary of the research and the results.

Keywords: keyword 1, keyword 2, keyword 3, keyword 4

1. Introduction

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2. Theory

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3. Simulation Setup

Simulations were performed using the VPIC (Vector Particle-In-Cell) code from LANL (Los Alamos National Laboratory). The Simulations were in 3D geometry ($N_x \times N_y \times N_z$) but we set y domain size to unity, e.g. $N_y = 1$. A shock wave is excited by the so-called *injection method*. A plasma consisting of electrons and ions is injected from the right-hand side boundary with an initial velocity $\boldsymbol{v}_0$ and travels toward to the negative x-direction. At the left-hand side boundary, particles are reflected by the conducting wall. Then, a shock forms due to the interactions between the incoming and the reflected particles, and it propagates in the x-direction with a shock speed v_{sh}. Therefore, the simulations are done in the downstream rest frame. The periodic boundary conditions are imposed in the y and z-direction.

Parameter units are normalized to natural PIC units: charges are in electron charge unit $e = 1$, masses are in electron mass unit $m_e = 1$, velocities are in speed of light unit $c = 1$, time is normalized by electron plasma frequency $\omega_{pe}^{-1} = 1$, length is normalized by electron inertial length $d_e = c\omega_{pe}^{-1}$ and permittivity of space $\epsilon_0 = 1$.

The domain spans from 0 to L_x in x, 0 to L_y in y and 0 to L_z in z. The resolution is $L_x/N_x = L_y/N_y = L_z/N_z = 0.1$. The mesh size (topology) is chosen such that the kinetic scale is well resolved. The number of particles per cell (nppc) is 100. The ion-electron mass ratio $m_i/m_e = 100$. The ratio of the plasma frequency to the electron cyclotron frequency is $\omega_{pe}/\Omega_{ce} = 5$. In the upstream frame, electron and ion have a temperature $T_0 = 0.01$ (in units of $m_e c^2$) and number density $n_0 = 1$. We use $B_0 = m_e c \Omega_{ce}/e$ for the upstream magnetic field

We consider three simulation cases that vary the shock normal angle θ (relative to the ambient magnetic field) and the injection velocity $\boldsymbol{v}_0$ as follows: (i) $\boldsymbol{v}_0 = 0.08c$ with $\theta = 75°$, (ii) $\boldsymbol{v}_0 = 0.16c$ with $\theta = 75°$, and (iii) $\boldsymbol{v}_0 = 0.16c$ with $\theta = 90°$.

Table 1: Simulation parameters.

Description	Parameter	Value
Domain x size	N_x	8192
Domain y size	N_y	1
Domain z size	N_z	1024
Simulation x size	L_x	$819.2d_e$
Simulation y size	L_y	$0.1d_e$
Simulation z size	L_z	$102.4d_e$
Simulation time	τ	$100\Omega_{ci}^{-1}$
Number of Particles per cell	nppc	100
Ion-to-electron mass ratio	m_i/m_e	100
Plasma-to-cyclotron ratio	ω_{pe}/Ω_{ce}	5
Initial plasma temperature	T_i, T_e	$0.01m_ec^2$
Initial number density	$n_{0,i}, n_{0,e}$	1
Upstream magnetic field	B_0	0.2
Ion inertial length	d_i	$0.1d_e$

TODO:

1. - beta parameters
2. - Alven speed
3. - Alven Mach number

4. Result

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5. Discussion

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Appendix A. Filtering functions.

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